

AutoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 1: Data and Model Selection

J.O. • Dr. David M. Simmonds • Evidence through September 28, 2026, 1:15 PM CDT

Decision and Evidence Boundary

The original September 24 evidence remains historical; later work below is a post-feedback supplement, not evidence that it was available at the original deadline. The old September 17 Muse adapter is retained only for further evaluation. Gemma development runs are not a factual-accuracy comparison. No model has been promoted and the independent final control has not been spent.

1. Data Characteristics

The task is a source-cited answer to a public Alabama Medicaid policy or provider-billing question, with a source locator or an abstention when support is absent. It is not an automated eligibility, payment, clinical or legal decision. Each candidate retains source URL, document identity, PDF page, snapshot date and SHA-256 file fingerprint. The target is a response grounded in the frozen source, not a class label.

Prepared Evidence	Measured Shape / Consequence
Source snapshot	138 public PDF files / 2,433 pages (September 14); 136 yielded candidate records.
Ordinary examples	1,999 answerable supervised fine-tuning (SFT) records; a separate one-record intentionally unanswerable case was quarantined.
Source mix	750 Administrative Code and 1,249 Provider Manual records. Median answer length 18 whitespace-delimited words; 95th percentile 52.
Historical clean split	1,684 train / 124 validation / 191 historical test records; 113 / 10 / 13 distinct source documents, respectively; zero document overlap.
Rights and privacy	Approved public information only; no patient claims, protected health information or agency-confidential records.

Preparation and Revised Split

The document-grouped split assigns every page of a source PDF together, preventing page-neighbor leakage. One blank-page safety example was removed from ordinary SFT. The September 16 mixed-generator run was rejected because 326 records came from a retired generator; it is procedural evidence only. September 18 used a different 1,685-row operational render and is not a strict repeat of the clean 1,684-row cohort.

The new experiment nests training subsets of 200, 509, 750 and 1,600 from the same 1,684 clean training rows, covers all 113 training documents even at 200, and holds the same 124 validation rows. Earlier proposals for separate 100/250/200/300-record configuration tests were deferred to protect one final test event. The already exposed historical 191 are retrospective diagnostics, not an untouched final test. The thirty hand-authored control questions remain locked; all their source documents are in the historical test group, none in train or validation. Their question wording and keys are independently authored, but their documents are not independent of the historical test documents.

2. Candidates Considered

The declared family is deep learning: local language-model adaptation. Instructor permission covers fine-tuning but not a weakened evaluation standard. The original two Muse runs are one approach plus a sensitivity run, not two architectures. The revised within-family alternatives differ in base architecture, memory demands, tokenizer and deployment feasibility; an eligible training smoke is not a model-quality result.

Approach	Why Plausible / Trade-Off	Verified Status
Muse Glimmer 30B, rank-16 low-rank adapter	Established local baseline; four-bit base and small trainable adapter reduce memory and retain rollback. Full merged deployment is not demonstrated.	Clean Sep. 17 run: 211 steps / 1 epoch. Sep. 18 is a different-render sensitivity run, not a distinct candidate.
google/gemma-4-26B-A4B-it, rank-4 adapter	Different mixture-of-experts architecture, smaller active footprint and explicit local training feasibility; tokenizer differs, so loss cannot rank it against Muse for factual answers.	Pinned exact revision; BF16 backward/resume smokes passed. N=200 and N=509 project-data runs completed; N=750 active at the reporting cutoff; N=1,600 not started.
nvidia/NVIDIA-Nemotron-3.5-Lightning-30B-A3B-BF16	Potential fast local serving, but license and exact architecture/hardware support require gates before training.	Considered; no verified project-data training or checkpoint gate.
ornith-ai/Ornith-1.5-35B-A3B	Alternative sparse model; local shared-GPU memory fit is uncertain. Do not displace another service just to test it.	Considered; inconclusive fit screen, no verified project-data training.

3. Baseline

The current-practice baseline is the student manually finding a source and writing an answer. On thirty simple, hand-authored code/page lookups, all thirty gold answers were source-validated; retrospectively paired session timestamps estimate 42:06 total (about 1:24 each). These are easy, non-blind lookups, not a stopwatch trial and not proof of time savings on policy reasoning. A second technical baseline is untouched Muse on the same frozen generated prompts as the two historical adapters; its human-scored pilot was 4/30. Neither baseline has yet been measured on the proposed harder control.

Why the Original Control Cannot Decide the Winner

All 30 items ask an answerable page or code lookup. Ten come from Jul26_34.pdf page 2, six from Jul26_28.pdf page 2 and five from Apr26_37.pdf page 2: 21/30 on three pages. This is an honest lookup diagnostic, not a representative policy-support or abstention benchmark. Eight to ten plausible unsupported cases and student-authored exception/date/conflict/multi-rule questions are required before an expanded control is frozen; none has been supplied or scored in this document.

4. Evidence and Evaluation

The committed primary outcome is human-adjudicated factual accuracy on answerable questions; unsupported-claim rate, correct locator and abstention are safety-critical secondary outcomes. Existing generated-set results are a small retrospective human pilot. No human-scored comparison on the independent hand-authored control or an expanded reasoning/abstention set has occurred. The table keeps different cohorts and metric types visibly separate rather than pretending they are one same-split ranking.

Condition / Cohort	Human Factual Accuracy	Train Fit	Validation Fit	Safety / Limits
Untouched Muse; historical generated pilot (n=30)	4/30 (13.3%)	N/A, no tuning	Not measured	Only 1/1 historical safety response; insufficient denominator.
Sep. 17 clean Muse; historical generated pilot (n=30)	11/30 (36.7%)	Loss 0.6201 (1,684 rows, 1 epoch)	Loss 0.5722 (124 rows)	0/1 historical abstentions; 4/191 outputs truncated; promotion gate failed.
Sep. 18 Muse sensitivity; historical generated pilot (n=30)	9/30 (30.0%)	Loss 0.6169 (1,685-row different render)	Loss 0.5736 (operational cohort)	Not a same-data architecture comparison.
Gemma, N=200 train; same 124 validation rows	Not scored	Epoch 3 loss 0.6893; next-token 84.40%	Epoch 3 loss 0.7044; next-token 83.45%	Train/validation only; 75 steps / 3 epochs.
Gemma, N=509 train; same 124 validation rows	Not scored	Epoch 5 loss 0.3373; next-token 90.81%	Best loss 0.6060 at epoch 2; epoch-5 loss 0.6636	Best checkpoint step 128; no final-control/test records used.
Gemma, N=750; provisional checkpoint at cutoff	Not scored	Epoch 2 loss 0.4791; next-token 87.69%	Epoch 2 loss 0.5829; next-token 85.58%	Checkpoint 188 only; training active, not complete.

Gemma uses BF16 rank-4 LoRA, learning rate 1×10^{-4} , assistant-response masking and a 1,024-token limit; the Muse clean run used four-bit rank-16 QLoRA, learning rate 2×10^{-4} and a 4,096-token limit. Next-token accuracy asks whether the model predicts the supplied next training token; it is not the percentage of policy questions answered correctly. Different model tokenizers and distinct training sizes forbid a numerical cross-family winner from those loss/accuracy columns. At N=509, validation loss worsened after epoch 2 while train fit improved; a resumed-run patience reset prevented the intended early-stop behavior. Select the best saved validation checkpoint, not the last checkpoint, for within-run analysis.

On the exposed 191-prompt historical generated set, Muse base versus Sep. 17: answer-token overlap F1 0.2076 versus 0.3973; numeric exact match 6.96% versus 17.39%; citation-URL presence 0% versus 24.08%. These are engineering diagnostics, NOT factual accuracy or verified source support. URL presence primarily shows learning the required output format. The predeclared 30-card human pilot is descriptive and may be optimistic because the generated answer key shares the Muse family; human/LLM rating agreement was not blinded. The formal promotion check returned pass=false because of truncation and a one-case abstention regression.

5. Selection and Justification

For continuation—not deployment—retain the Sep. 17 Muse adapter as the historical provisional condition, and continue the separately identified Gemma training-size experiment. The Sep. 17 pilot exceeds untouched Muse on the same thirty generated questions, but the difference cannot establish performance on realistic questions. Gemma improves the range of model choices yet lacks a comparable human-scored factual outcome. Nemotron and Ornith remain feasibility candidates. A definitive cross-family winner is withheld: the authorized train/validation measures are tokenizer-specific next-token metrics, and the protected control cannot be used to choose a winner after looking at its answers.

6. Interpretability and Leakage Check

The practical interpretability check for a language model is whether answers depend on the right exception, effective date, negation and source locator—not merely whether it emits a URL. No paired ablation has been run. Structural checks found zero document overlap, two duplicate-question groups with no cross-split match, one train-validation near-duplicate question candidate at Jaccard ≥ 0.85 , and sixteen cross-split duplicate-answer groups affecting fifty-seven records. The group/record counts are reproduced from the canonical file with the project's punctuation-insensitive normalization; Appendix C lists each group. Identical short codes may be coincidental; repeated substantive policy text or template language may transmit information across split boundaries. Neither possibility is a human disposition.

Open Leakage and Quality Gate

The student must review all 16 answer groups against both questions and source pages, decide which are harmless shared values, boilerplate, or substantive repeated content, and remove/re-group any contaminated held-out item before a clean comparative score. The near-duplicate question pair also needs a source-level ruling. Existing LLM-scored workbook examples are not a human sign-off. A flagged H2011 “Daily” table-column error requires human verification and correction/quarantine before any affected row is reused.

Every existing control item maps by full source URL and SHA-256 to a historical test document; Appendix B makes the per-item result visible. That refutes the specific concern that these thirty questions draw on training documents in the clean lineage, but does not fix their simple wording, source-page concentration, possible generated-test document overlap or absence of unanswerable cases. A revised student-authored control should be frozen separately and excluded from training, development tuning and candidate choice. Plan a sufficiently varied, source-verified answerable stratum and eight to ten plausible-but-unsupported items; score abstention and locator correctness with explicit denominators and a human adjudication key. Do not call the old 1/1 safety probe a 90% estimate.

Attribution Is a Later Experiment

Open WebUI and ChromaDB retrieval have not been built and deployed for this comparison. No retrieval benefit is claimed. Model-only control scoring can follow a legitimate selection and dataset lock; paired retrieval-on versus retrieval-off attribution waits for the later deployed user interface. This dependency does not excuse unresolved leakage or missing model-only policy-support evidence.

7. What Would Change My Mind

A candidate would be rejected or reconsidered if independently scored, source-verified policy questions do not improve answerable factual accuracy by at least 10 percentage points against untouched base, locator correctness is under 95%, unsupported claims reach 2% or more, abstention is under 90% on a meaningful unsupported set, safety routing is under 95%, general/format regression exceeds 3 points, interactive latency fails, or any leakage or rights defect persists. These are project targets, not results. A strong next-token score or a properly formatted URL cannot override human-scored factual and safety failures.

The October 9, 5:00 PM CDT fallback remains a decision gate: if an adapter-loaded or merged model has not produced a complete validated answer, stop adaptation on the critical path and pursue untouched base plus versioned retrieval, reporting adaptation as a negative result. If retrieval later supplies nearly all of the benefit, credit retrieval rather than fine-tuning.

8. Repository and Reproducibility

Public course deliverables: github.com/porespellar/INAI6986_project_deliverables

Direct executed historical notebook: [milestone1_reproducibility.ipynb](#)

The existing executed notebook rebuilds the historical 1,999-record ordinary dataset from the frozen candidates, verifies source hashes, reproduces the 1,684/124/191 split and earlier diagnostics from saved outputs. It does not reproduce new Gemma training or a not-yet-existent independent-control score. The local experiment manifests, run summaries and tested split-rendering scripts are listed in Appendix D; they must be incorporated into an executed, portable notebook before any new metric is represented as notebook-reproducible to the instructor. The public repo and notebook above are historical links, not a claim that this revision has been uploaded.

Evidence Gates Before Comparative Scoring

Gate	Current Disposition / Required Action
Source membership	Closed for original 30: 30/30 on historical test documents; zero on clean train/validation. Preserve hash-matched item mapping.
Question validity	Open: student authors exception, effective-date, conflict and multi-rule items across more sources; original 30 remain lookup-only.
Abstention	Open: add 8-10 independently authored unsupported questions and source-verify the bounded absence of an answer.
Duplicate/quality flags	Open: human-dispose all 16 groups / 57 records and one near-duplicate question pair; resolve flagged H2011 row.
Candidate choice	Open: train/validation token metrics alone cannot rank cross-tokenizer factual generalization; do not peek at protected control.
Final model test	Not run: lock dataset, recipe, checkpoint, comparator and scorer first; evaluate the frozen candidate and untouched base once.

This supplement addresses the feedback with verified repairs and an explicit unfinished-work boundary. It is not a claim that the expanded control, human leakage disposition, final comparison, or a particular course score has been achieved.

Appendix A — AI Use Disclosure

AI assistance. Hermes Agent (GPT-6 Sol 900k) reviewed the professor's feedback, rubric, saved project evidence and document layout; wrote and validated this supplement and its build code. Earlier local Muse generated the original SFT candidates. AI-suggested leakage categories and document prose are not human source adjudication, independently authored control questions, or model outcomes.

Human authorship and accountability. J.O. conceived the project and selected the architecture, source scope, model candidates, metrics, split and guarded workflow; configured local hardware and services, directed agents, diagnosed training and checkpoint failures, and made proceed/stop/quarantine decisions. J.O. wrote and source-verified the original thirty control questions and answer keys. J.O. must author any new control questions, review flagged source evidence and approve final conclusions before submission. The student retains final responsibility for all content and project actions.

Verification. The builder reads fixed local evidence, asserts the clean dataset hash, the 30 historical test-document mappings and the 16-group/57-record duplicate count, and checks new Gemma summary identities. DOCX package validation, PDF text readback and page-level visual review are delivery checks, not substitutes for human adjudication.

Appendix B — Independent Control Source-Split Map

The original source-verified, student-authored set contains thirty answerable lookup questions from ten PDF documents and eleven document/page locations. The three most concentrated pages account for 21/30.

Mapping uses full source URL plus SHA-256 against the canonical f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c dataset; every listed source is in the historical test split. The original questions are preserved verbatim; their source screenshot records remain in the original submission and local evidence directory. The table is a membership audit, not an expanded control or a model score.

ID	Original Student-Authored Question	Expected Answer	Source PDF / Page	Split
SA-001	What page number in the table of contents lists Rule 560-X-1-.07, Provider Rights and Responsibilities?	Page 3.	9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1	test
SA-002	What page number in the table of contents lists Rule 560-X-28-.02, Summary of State Law Regarding Advanced Directives?	Page 2.	9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1	test
SA-003	What page number in the table of contents lists Rule 560-X-49-.07, Copayment?	Page 2.	9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1	test
SA-004	What page number in the table of contents lists Rule 560-X-50-.03, Recipient Eligibility?	Page 3.	9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1	test
SA-005	What Medicaid provider type is assigned to Licensed Social Workers?	Provider type 11, Crossovers Only.	Apr26_23.pdf p.1	test
SA-006	What provider type is assigned to therapy providers?	Provider type 17, Therapy.	Apr26_37.pdf p.2	test
SA-007	What provider type is assigned to Remote Patient Monitoring providers?	Provider type 08, Remote Patient Monitoring.	Jul26_111.pdf p.1	test
SA-008	What provider type is assigned to physicians?	Provider type 31, Physician.	Jul26_28.pdf p.2	test
SA-009	What provider type is assigned to preventive health education providers?	Provider type 55.	Jul26_30.pdf p.1	test
SA-010	What provider type is assigned to psychology providers?	Provider type 54, Psychologist.	Jul26_34.pdf p.2	test
SA-011	What provider type is assigned to Behavioral Health providers?	Provider type 07, Behavioral Health.	Jul26_34.pdf p.2	test
SA-012	What specialty code is listed for Psychology providers?	Psychology specialty code 112.	Jul26_34.pdf p.2	test
SA-013	What specialty code is listed for a Licensed Professional Counselor / LPC?	Specialty code 070	Jul26_34.pdf p.2	test
SA-014	What provider type is assigned to Nurse Practitioners?	Provider type 09.	Jul26_28.pdf p.2	test
SA-015	What provider type is assigned to Physician Assistants?	Provider type 10.	Jul26_28.pdf p.2	test

Control Source-Split Map — Continued

ID	Original Student-Authored Question	Expected Answer	Source PDF / Page	Split
SA-016	What specialty code is listed for Occupational Therapy?	Occupational Therapy code 171.	Apr26_37.pdf p.2	test
SA-017	What specialty code is listed for Physical Therapy?	Physical Therapy code 170.	Apr26_37.pdf p.2	test
SA-018	What specialty code is listed for Speech Therapy?	Speech Therapy code 173.	Apr26_37.pdf p.2	test
SA-019	What specialty code is listed for Applied Behavior Analysis Therapy.	Applied Behavior Analysis (ABA) Therapy. Code 175.	Apr26_37.pdf p.2	test
SA-020	What provider specialties are assigned to Remote Patient Monitoring providers?	Provider specialties 085 or 086.	Jul26_111.pdf p.1	test
SA-021	What specialty code is listed for an Associate Licensed Counselor / ALC?	Specialty code 071.	Jul26_34.pdf p.2	test
SA-022	What specialty code is listed for a Licensed Marriage and Family Therapist / LMFT?	Specialty code 072.	Jul26_34.pdf p.2	test
SA-023	What specialty code is listed for a Licensed Marriage and Family Therapist Associate / LMFTA?	Specialty code 075.	Jul26_34.pdf p.2	test
SA-024	What specialty code is listed for a Licensed Master Social Worker / LMSW?	Specialty code 073.	Jul26_34.pdf p.2	test
SA-025	What specialty code is listed for a Licensed Independent Clinical Social Worker / LICSW.	Specialty code 074.	Jul26_34.pdf p.2	test
SA-026	What specialty code is listed for a Licensed Psychological Technician / LPT?	Specialty code 076.	Jul26_34.pdf p.2	test
SA-027	What specialty code is listed for Preventive Health Education?	Specialty code 183.	Jul26_30.pdf p.1	test
SA-028	What specialty code is listed for EPSDT in the physician specialty table?	EPSDT code 560.	Jul26_28.pdf p.2	test
SA-029	What specialty code is listed for Family Practice in the physician specialty table?	Family Practice code 316.	Jul26_28.pdf p.2	test
SA-030	What specialty code is listed for Cardiac Surgery in the physician specialty table?	Cardiac Surgery code 312.	Jul26_28.pdf p.2	test

Source concentration (document/page → number of questions): Jul26_34.pdf p.2: 10; Jul26_28.pdf p.2: 6; Apr26_37.pdf p.2: 5; Jul26_111.pdf p.1: 2; Jul26_30.pdf p.1: 2; 9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1: 1; 9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1: 1; 9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1: 1; 9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1: 1; Apr26_23.pdf p.1: 1.

Appendix C — Cross-Split Answer-Overlap Audit

These are reproducible flags, not human rulings. The canonical file yields sixteen punctuation/case-normalized answer groups spanning splits (57 member records). Review the questions, cited source pages and semantic meaning for every group in the human review workbook; a shared page number, repeated boilerplate and repeated policy answer require different remedies. Do not delete records merely because an answer string repeats. Keep the frozen original file and version a corrected split/file after human approval.

Answer Group (abbreviated)	Members	Splits	Provisional Risk / Action
page 2	8	test/train/validation	Needs source-level human ruling
this page intentionally left blank	7	test/train/validation	Needs source-level human ruling
the current procedural terminology cpt and current dental termin...	6	train/validation	Needs source-level human ruling
page 3	5	test/train/validation	Needs source-level human ruling
3 90	4	test/train	Needs source-level human ruling
icd 9 codes should be used for claims submitted with dates of se...	4	test/train/validation	Needs source-level human ruling
3	3	test/train	Needs source-level human ruling
4	3	train/validation	Needs source-level human ruling
chapter 5	3	train/validation	Needs source-level human ruling
11	2	test/train	Needs source-level human ruling
334 215 7416	2	test/train	Needs source-level human ruling
a hard copy ub 04 claim form must be submitted	2	train/validation	Needs source-level human ruling
the amount billed shall not exceed the entity s actual acquisiti...	2	test/train	Needs source-level human ruling
the current procedural terminology cpt and current dental termin...	2	train/validation	Needs source-level human ruling
the ltc gainwell cover sheet from the web portal must accompany ...	2	train/validation	Needs source-level human ruling
up to four procedure code modifiers	2	test/train	Needs source-level human ruling

The one cross-split near-duplicate question pair remains open. The separate quality-review human template and AI example are in artifacts/milestone1/quality_review/2026-09-24/; no sign-off is inferred from their existence. The original generated-set thirty-card human factual pilot is a different activity.

Appendix D — Local Evidence Ledger and Sources

Claim / Requirement	Verifiable Artifact
Rubric and instructor's scored feedback	Assignment 3: Milestone 1 — Data & Model Selection, course rubric; instructor's scored feedback (private course file)
Historical notebook and original course record	notebook/milestone1_reproducibility.ipynb; artifacts/milestone1/2026-09-22/{progress_snapshot,evaluation_snapshot}.json; original v2 PDF
Clean dataset and frozen control	artifacts/milestone1/2026-09-17/approved_candidates.jsonl (SHA-256 f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c); data/curated/student-authored-control-set-2026-09-22.jsonl (SHA-256 a1deb52304d7bce939593a228836debe2359488159e6e3b112e9d48ac7bea55c)
Human pilot and quality flags	artifacts/milestone1/human_scoring/2026-09-24/human_generated_sample_scores.json; artifacts/milestone1/quality_review/2026-09-24/INAI6986_1999Record_QualityReview_HumanTemplate.xlsx
Nested train sets and Gemma rendering	artifacts/model_comparison/2026-09-25/{nested-train-subset-manifest,gemma-render-validation,source-state-inventory}.json; scripts/{prepare_comparison_splits,prepare_gemma_comparison_splits,render_gemma_sft_data}.py
Gemma N=200 and N=509 actual results	artifacts/model_comparison/2026-09-25/training_results/gemma4-{n200-training-summary,n509-resumed-training-summary}.json; results.jsonl; run_log.csv
N=750 provisional checkpoint and untouched final control	artifacts/model_comparison/2026-09-25/training_results/gemma4-n750-checkpoint188-trainer-state.json; N=750 live read-only observation at 2026-09-28 13:15 CDT; no final control scores

Selected Technical Sources

Alabama Medicaid Administrative Code:

https://medicaid.alabama.gov/content/9.0_Resources/9.2_Administrative_Code.aspx

Alabama Medicaid Provider Manuals:

https://medicaid.alabama.gov/content/Gated/7.6.1G_Provider_Manuals.aspx

LoRA method: Hu et al., <https://arxiv.org/abs/2106.09685>. Candidate model identities and exact pinned revisions are recorded in artifacts/model_comparison/2026-09-25/source-state-inventory.json; no third-party performance claim is inferred from a model card.